

DATA ANALYTICS IN PYTHON

MODULE 1 - PYTHON FOUNDATIONS

Question Bank & Practical Problem Set

University of Agriculture Faisalabad · Grads/Undergrads · 2026

150 MCQs	15 Practical Problems	6 Topics	70/30 Exam Split
Section A	Section B	Weeks 1-2	Seen + Unseen

SECTION A - MULTIPLE CHOICE QUESTIONS

150 Questions · 6 Topics · ~25 per topic
Difficulty: 40% Low · 40% Medium · 20% High

Instructions: Each question has exactly one correct answer. Circle or underline your choice.

TOPIC 1: VARIABLES, DATA TYPES & OPERATORS (Q1–Q25)

Q1. What does `type(True).__name__` return?

- a) "boolean" b) "bool" c) "int" d) "True"

Q2. What is the output of `10 // 3`?

- a) 3.33 b) 3.0 c) 3 d) 4

Q3. Which expression evaluates to True?

- a) `0.1 + 0.2 == 0.3` b) `bool(0)` c) `isinstance(True, int)` d) `type(True) == int`

Q4. What does `print(2 ** 3 ** 2)` output?

- a) 64 b) 512 c) 36 d) 6

Q5. What does `int("3.7")` raise or return?

- a) 3 b) 4 c) 3.7 d) ValueError

Q6. What does `bool("")` return?

- a) True b) False c) None d) TypeError

Q7. What is the output of `17 % 5`?

- a) 2 b) 3 c) 3.4 d) 2.0

Q8. What does `a = b = 5` do in Python?

- a) Raises SyntaxError b) Assigns 5 to b then b to a c) Both a and b reference the same integer object 5 d) Creates two separate integer objects

Q9. What is x after: `x = 5; x += 3; x *= 2`?

- a) 11 b) 16 c) 13 d) 26

Q10. What does `round(2.675, 2)` return in CPython?

- a) 2.68 b) 2.67 c) 2.675 d) 2.7

Q11. Which is a valid Python variable name?

- a) 2data b) data-set c) _data_2 d) data set

Q12. What does `print(type(None).__name__)` output?

- a) "none" b) "None" c) "NoneType" d) "null"

Q13. What does `5 / 2` return in Python 3?

- a) 2 b) 2.5 c) 2.0 d) Error

Q14. Which correctly checks if `x` is int OR float?

- a) `type(x)==float or int` b) `isinstance(x,(int,float))` c) `type(x) in [int, float]` d) Both B and C

Q15. What does `f"{3.14159:.2f}"` produce?

- a) "3.14" b) "3.14159" c) "3.142" d) Error

Q16. What does `not not 0` evaluate to?

- a) True b) False c) 0 d) 1

Q17. What is the output of `True + True + False`?

- a) Error b) True c) 2 d) 1

Q18. Which statement about Python integers is correct?

- a) Limited to 64 bits b) Can be arbitrarily large c) Overflow raises `OverflowError` d) Large integers auto-convert to float

Q19. What does `id(x) == id(y)` being True confirm?

- a) `x` and `y` have the same value b) `x` and `y` are the same object in memory c) `x` and `y` have the same type d) `x == y` is always True

Q20. What is operator precedence? (highest first)

- a) `+` `-`, `*` `//`, `**`, comparisons b) `**`, `*` `//`, `+` `-`, comparisons c) comparisons, `**`, `*` `//`, `+` `-` d) `*` `//`, `**`, comparisons, `+` `-`

Q21. What does `int(True) + int(False) + int(None)` raise?

- a) 1 b) 0 c) `TypeError` d) 2

Q22. What does `"5" + 5` produce in Python 3?

- a) "55" b) 10 c) `TypeError` d) "10"

Q23. What does `divmod(13, 4)` return?

- a) (3, 1) b) (3.25, 1) c) (3, 1.0) d) 3.25

Q24. Which expression evaluates to a float?

- a) `10 / 2` b) `10 // 2` c) `int(10.5)` d) `2 ** 3`

Q25. What does `print(0.1 + 0.2 == 0.3)` output?

a) True - they are exactly equal b) False - floating-point binary precision issues c) Error - cannot compare floats with == d) True - Python rounds the result

ANSWER KEY

Q#	Ans	Key Concept / Explanation
1	B	bool is the class name. Note that <code>isinstance(True, int)</code> is also True since bool is a subclass of int.
2	C	Floor division (<code>//</code>) returns the integer quotient, truncating toward negative infinity.
3	C	bool is a subclass of int in Python, so <code>isinstance(True, int)</code> is True. <code>0.1+0.2</code> has floating-point precision issues.
4	B	<code>**</code> is right-associative: <code>3**2 = 9</code> first, then <code>2**9 = 512</code> . Not <code>(2**3)**2 = 64</code> .
5	D	<code>int()</code> cannot directly parse a decimal string. Use <code>int(float("3.7"))</code> to get 3.
6	B	Empty strings, 0, None, empty lists/dicts/sets are all falsy in Python.
7	A	$17 = 5 \times 3 + 2$. The <code>%</code> operator returns the remainder: $17 - 5 \times 3 = 2$.
8	C	Chained assignment binds both names to the same object. <code>id(a) == id(b)</code> will be True.
9	B	$5 + 3 = 8$, then $8 * 2 = 16$.
10	B	Floating-point binary representation makes 2.675 slightly less than 2.675 internally, so it rounds down to 2.67.
11	C	Variable names must start with a letter or underscore. Hyphens and spaces are not allowed.
12	C	None's type is <code>NoneType</code> , not None itself. <code>type(None)</code> returns <code><class 'NoneType'></code> .
13	B	Python 3 uses true division for <code>/</code> . $5/2 = 2.5$ (float). Use <code>5//2</code> for integer division.

14	D	isinstance() supports tuple of types. type(x) in [int,float] also works. Never use type(x)==float or int - the or int always evaluates truthy.
15	A	.2f means 2 decimal places, so 3.14159 rounds to 3.14.
16	B	not 0 = True (0 is falsy), then not True = False.
17	C	bool is a subclass of int: True=1, False=0. So 1+1+0 = 2 (integer).
18	B	Python integers have arbitrary precision - 2**1000 works without overflow.
19	B	id() returns the memory address. Equal ids mean same object (identity, not equality).
20	B	PEMDAS in Python: ** > * // % > + - > comparisons. Remember: 2+3*4 = 14, not 20.
21	C	int(None) raises TypeError. int(True)=1 and int(False)=0 are valid, but None cannot be converted.
22	C	You cannot concatenate str and int. Use int("5") + 5 or "5" + str(5).
23	A	divmod(a,b) returns (a//b, a%b): quotient and remainder. 13//4=3, 13%4=1.
24	A	/ always returns float in Python 3. // returns int (if both operands are int). ** returns int.
25	B	0.1+0.2 in binary ≈ 0.30000000000000004. Always use math.isclose() for float comparisons.
26	B	[::-1] reverses the string. P=0,Y=1,T=2,H=3,O=4,N=5 reversed = NOHTYP.
27	A	split() with no argument splits on any whitespace and removes extra